Provably Powerful Graph Neural Networks for Directed Multigraph

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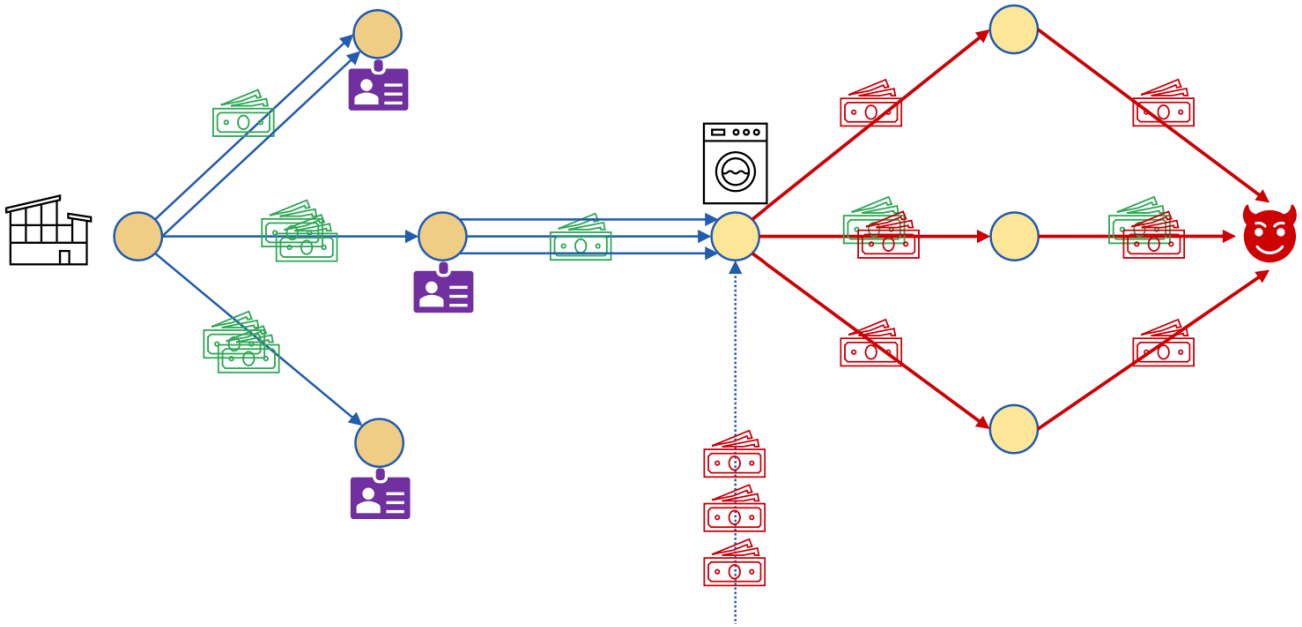
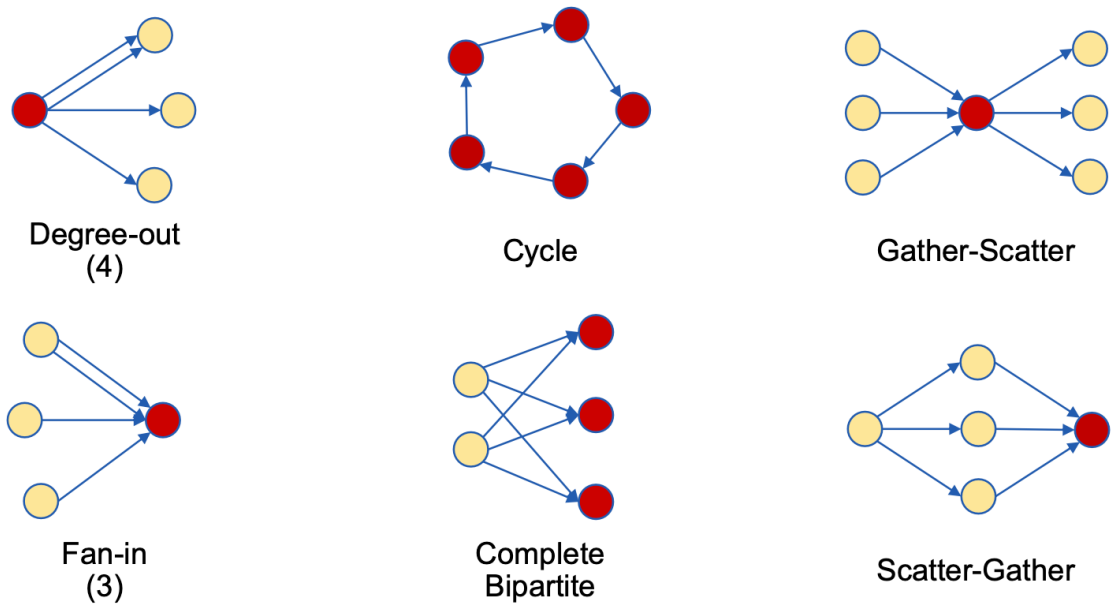
Money Laundering Detection



Directed MultiGraph

Subgraph Detection

Standard GNNs can not detect the subgraph patterns!



Suspicious Patterns

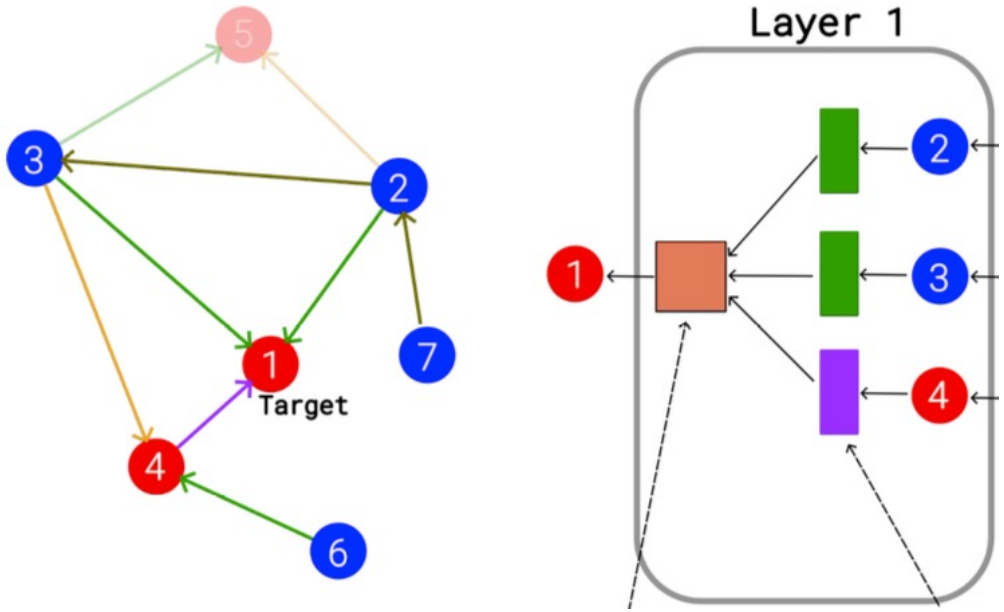
Directed MultiGraphs

Ranging from chemistry to social networks

Message Passing GNNs (MPNNs)

Node v
Account Features: $h^0{}_{(v)}$
Features: Account Number, Bank ID, account balance

Edge e = (u,v)
Transaction Features: $h^0{}_{(u,v)}$
Features: Amount, currency, timestamp



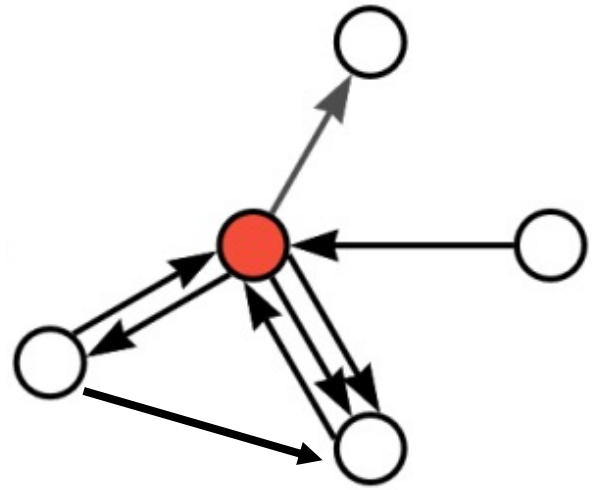
MPNNS
GCN
GAT
GIN

$$a^{(t)}(v) = \textsc{Agg}\left(\{(h^{(t-1)}(u), h^{(0)}((u,v))) \mid u \in N_{in}(v)\}\right)$$

$$h^{(t)}(v) = \textsc{Update}\left(h^{(t-1)}(v), a^{(t)}(v)\right),$$

Limitations in Standart MPNNs

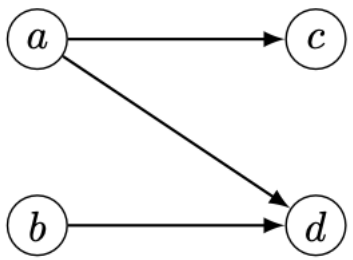
- Can not see outgoing edges
- Can not distinguish neighbours
- Can not detect cycles



Methods

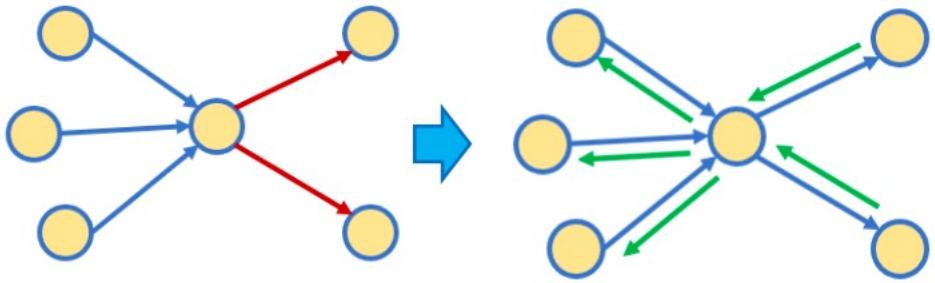
1. Reverse Message Passing (Jaume et al. 2019)
2. Port Numbering (Sato, Yamada, and Kashima 2019)
3. Ego IDs (You et al. 2021)

Reverse Message Passing



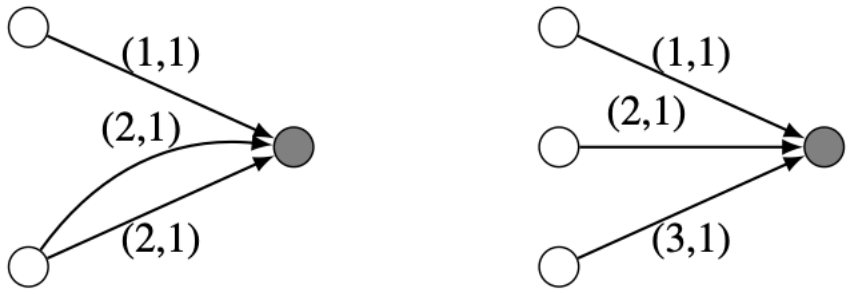
Nodes (a and b) with different out-degrees are not distinguishable

$$a_{in}^{(t)}(v) = \textsc{Agg}_{in}\left(\{h^{(t-1)}(u) \mid u \in N_{in}(v)\}\right),$$
$$a_{out}^{(t)}(v) = \textsc{Agg}_{out}\left(\{h^{(t-1)}(u) \mid u \in N_{out}(v)\}\right),$$
$$h^{(t)}(v) = \textsc{Update}\left(h^{(t-1)}(v), a_{in}^{(t)}(v), a_{out}^{(t)}(v)\right),$$



Proposition 1. An MPNN with sum aggregation and reverse MP can solve degree-out.

Directed Multigraph Port Numbering



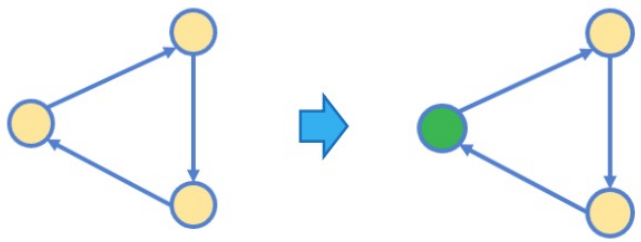
Needs to distinguishes the edges from the same neighbor and edges from different neighbors

Proposition 2. An MPNN with max aggregation and multigraph port numbering can solve fan-in.

Proposition 3. An MPNN with max aggregation, multigraph port numbering, and reverse MP can solve fan-out.

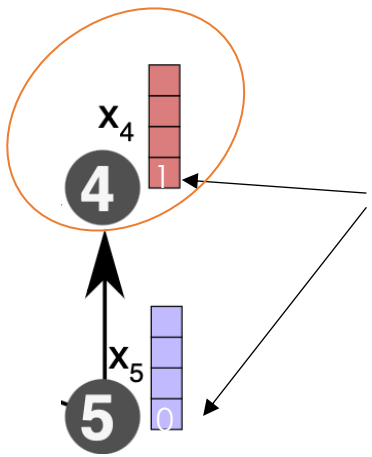
Ego IDs

Cycles & Ego IDs



Marked node can recognize when a sequence of messages cycles back around to it

Mark the center nodes of the batch!

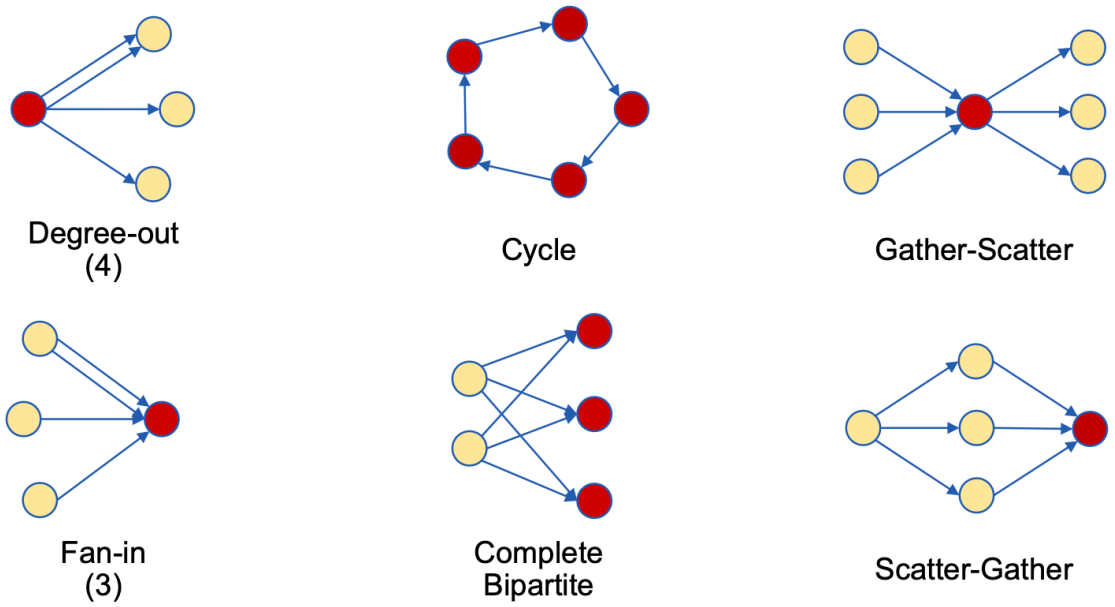


Ego IDs feature added

Corollary 5. GIN with ego IDs, multigraph port numbering, and reverse MP can theoretically detect any directed subgraph pattern.

Adapt to MultiGraph

Base GNN + ReverseMP +
Ports + EgoIds (MultiGNN)



Experiments

Datasets:

- Synthetic Pattern Detection Tasks.
- Anti-Money Laundering (AML) (Altman et al. 2023)
- Ethereum Phishing Detection (ETH) (Chen et al. 2019a)
- Real-World Directed Graph Datasets

Scoring:

- F1 (due to very imbalanced datasets)

Base GNN Models:

- GIN with edge features (Hu et al. 2019)
- GAT (Velickovic et al. 2017)
- PNA (Velickovic et al. 2019)
- R-GCN (Schlichtkrull et al. 2018)

Tree Based Models:

- XGBoost (Chen and Guestrin 2016) with pre-calculated graph- based features (GFs)
- LightGBM (Ke et al. 2017) with pre-calculated graph- based features (GFs)

Results

Synthetic Pattern Detection Tasks

Model	deg-in	deg-out	fan-in	fan-out	C2	C3	C4	C5	C6	S-G	B-C
GIN (Xu et al. 2018; Hu et al. 2019)	99.77	43.58	95.57	35.91	34.67	58.00	50.80	43.12	48.59	69.31	63.12
GAT (Veličković et al. 2017)	10.33	10.53	9.69	0.00	0.00	0.00	25.86	0.00	0.00	0.00	0.00
PNA (Velickovic et al. 2019)	99.63	43.02	95.00	38.93	25.77	54.75	51.92	48.79	48.40	65.88	65.51
GIN+EU (Battaglia et al. 2018)	99.30	42.74	95.70	39.13	32.58	55.91	54.65	47.62	49.68	68.54	64.64
GIN+EgoIDs (You et al. 2021)	99.78	51.48	95.06	49.24	98.13	97.97	53.12	44.37	45.42	66.44	63.90
GIN+Ports (Sato, Yamada, and Kashima 2019)	99.47	45.00	99.59	41.51	27.79	56.11	42.68	41.11	44.99	67.99	65.76
GIN+ReverseMP (Jaume et al. 2019)	98.87	99.08	94.99	95.25	35.96	63.85	69.09	67.44	71.23	65.83	66.18
+Ports	98.41	98.35	98.51	99.16	39.15	63.58	69.00	70.35	75.04	67.84	65.78
+EgoIDs (Multi-GIN)	99.48	99.09	99.62	99.32	98.97	98.73	97.46	91.60	84.23	97.42	94.33
Multi-GAT	98.68	98.36	99.28	99.33	98.61	98.93	98.90	95.82	91.81	96.66	86.92
Multi-PNA	99.64	99.25	99.53	99.41	99.71	99.54	99.49	97.46	88.75	99.07	96.77
Multi-GIN+EU	99.55	99.53	99.76	99.77	99.37	99.71	98.73	95.73	88.38	98.81	97.82

The Ck abbreviations stand for directed k-cycle detection, S-G stands for scatter-gather and B-C stands for biclique detection.

Real-World Directed Graph Datasets

Model	Squirrel	Chameleon	Arxiv-Year
Grad. Gating (SOTA)	$\mathbf{64.26 \pm 2.38}$	71.40 ± 2.38	63.30 ± 1.84
GIN (GCN*)	$35.94 \pm 2.01^*$	51.29 ± 0.63	50.15 ± 0.26
+ReverseMP	$59.84 \pm 2.08^*$	66.36 ± 1.23	63.31 ± 0.45
+Ports	$59.33 \pm 2.40^*$	73.46 ± 0.97	$\mathbf{68.12 \pm 0.51}$
+EgoIDs	$58.41 \pm 2.49^*$	$\mathbf{73.73 \pm 1.03}$	68.04 ± 0.60

Results

AML and ETH Dataset

Model	AML Small HI	AML Small LI	AML Medium HI	AML Medium LI	ETH
LightGBM+GFs (Altman et al. 2023)	62.86 ± 0.25	20.83 ± 1.50	59.48 ± 0.15	20.85 ± 0.38	53.20 ± 0.60
XGBoost+GFs (Altman et al. 2023)	63.23 ± 0.17	27.30 ± 0.33	65.70 ± 0.26	28.16 ± 0.14	49.40 ± 0.54
GIN (Xu et al. 2018; Hu et al. 2019)	28.70 ± 1.13	7.90 ± 2.78	42.20 ± 0.44	3.86 ± 3.62	26.92 ± 7.52
PNA (Velickovic et al. 2019)	56.77 ± 2.41	14.85 ± 1.46	59.71 ± 1.91	27.73 ± 1.65	51.49 ± 4.26
GIN+EU (Battaglia et al. 2018)	47.73 ± 7.86	20.62 ± 2.41	49.26 ± 4.02	6.19 ± 8.32	33.92 ± 7.34
R-GCN (Schlichtkrull et al. 2018)	41.78 ± 0.48	7.43 ± 0.38	OOM	OOM	OOM
GIN+EgoIDs (You et al. 2021)	39.65 ± 4.73	14.98 ± 2.66	45.26 ± 2.16	11.17 ± 6.41	26.01 ± 2.27
GIN+Ports (Sato, Yamada, and Kashima 2019)	54.85 ± 0.89	21.41 ± 2.40	54.22 ± 1.94	10.51 ± 12.82	32.96 ± 0.25
GIN+ReverseMP (Jaume et al. 2019)	46.79 ± 4.97	15.98 ± 4.39	51.93 ± 2.90	14.00 ± 9.34	36.86 ± 8.12
+Ports	56.85 ± 2.64	23.80 ± 4.07	57.15 ± 0.76	11.39 ± 8.36	42.51 ± 7.16
+EgoIDs (Multi-GIN)	57.15 ± 4.99	22.12 ± 2.88	56.23 ± 1.51	14.55 ± 2.91	42.86 ± 2.53
Multi-GIN+EU	64.79 ± 1.22	26.88 ± 6.63	58.92 ± 1.83	16.30 ± 4.73	48.37 ± 6.62
Multi-PNA	64.59 ± 3.60	30.65 ± 2.00	65.67 ± 2.66	33.23 ± 1.31	65.28 ± 2.89
Multi-PNA+EU	68.16 ± 2.65	33.07 ± 2.63	66.48 ± 1.63	36.07 ± 1.17	66.58 ± 1.60

HI indicates a higher illicit ratio and LI indicates a lower illicit ratio.

Results aimed at reproduce and improve

Dataset:

- Anti-Money Laundering (AML) (Altman et al. 2023) Small HI

Model	AML Small HI
LightGBM+GFs (Altman et al. 2023)	62.86 ± 0.25
XGBoost+GFs (Altman et al. 2023)	63.23 ± 0.17
GIN (Xu et al. 2018; Hu et al. 2019)	28.70 ± 1.13
PNA (Velickovic et al. 2019)	56.77 ± 2.41
GIN+EU (Battaglia et al. 2018)	47.73 ± 7.86
R-GCN (Schlichtkrull et al. 2018)	41.78 ± 0.48
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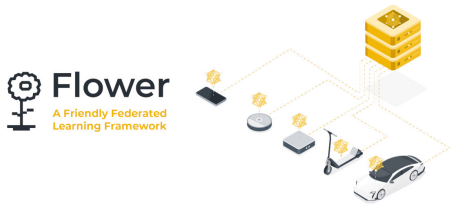
GIN+ReverseMP (Jaume et al. 2019)	46.79 ± 4.97
+Ports	56.85 ± 2.64
+EgoIDs (Multi-GIN)	57.15 ± 4.99
Multi-GIN+EU	64.79 ± 1.22
Multi-PNA	64.59 ± 3.60
Multi-PNA+EU	68.16 ± 2.65

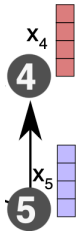
Aimed Improvement

- Applying Federated Learning

or

- Extending meaningful features with methods in literature such as network centrality





Thank You!

Questions?